

AMAZON BEAUTY RECOMMENDATION SYSTEM

Comparative Evaluation of Popularity-Based, Item-Based Collaborative Filtering, and SVD Models

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KEY FINDING: SVD DELIVERS THE STRONGEST BALANCE BETWEEN PREDICTION ACCURACY, RANKING QUALITY, AND PERSONALIZATION.

1. Research Problem

🇺🇸 **99.50% Sparsity:** Limits personalization quality.

❓ **Cold-Start:** New items lack interaction data.

"Can advanced models outperform the popularity baseline?"

Objective

Identify the model providing the strongest balance between prediction accuracy, ranking quality, personalization, and coverage.

Hypotheses

🎯 H1

Matrix factorization improves accuracy.

🔍 H2

Popularity helps cold-start users.

📊 H3

RMSE alone isn't sufficient for

🕒 H4

Activity explains model behavior.

2. Dataset Overview

50,000

RATINGS

4,999

USERS

1,999

PRODUCTS

99.50%

SPARSITY

1-5

RATING SCALE

Preprocessing Summary



Validation

Types & Nulls



Timestamp

Conversion



Features

Cold-Start/Pop



Split

80/20 Train-Test

3. Analytical Approach



Dataset



Preparation

Popularity

Item-CF

SVD

Model



Evaluation

Model Rationale

Popularity: Benchmark/Cold-start. **Item-CF:** Reach/Catalog reach. **SVD:** Latent preference capture.

Why SVD?

Captures latent preference patterns and excels in sparse environments (99.50%) through low-rank decomposition, enabling superior personalization.

4. Accuracy vs Coverage: The Strategic Frontier



"SVD achieves the strongest balance between recommendation accuracy, ranking quality, personalization, and practical recommendation coverage."

Strategic Implications

Recommendation-system design directly influences product discovery, engagement, and business performance. SVD provides the strongest balance between predictive accuracy and recommendation quality.

Decision Recommendation

Deploy SVD Matrix Factorization as the primary engine, while using Item-CF for catalog exploration and Popularity for cold-start scenarios.

5. Performance Summary

Model	RMSE	NDCG@10	Coverage
Popularity Baseline	1.44	0.0025	0.60%
Item-Based CF	1.68	0.0018	99.85%
SVD Matrix Factorization	1.38★	0.0030★	22.01%

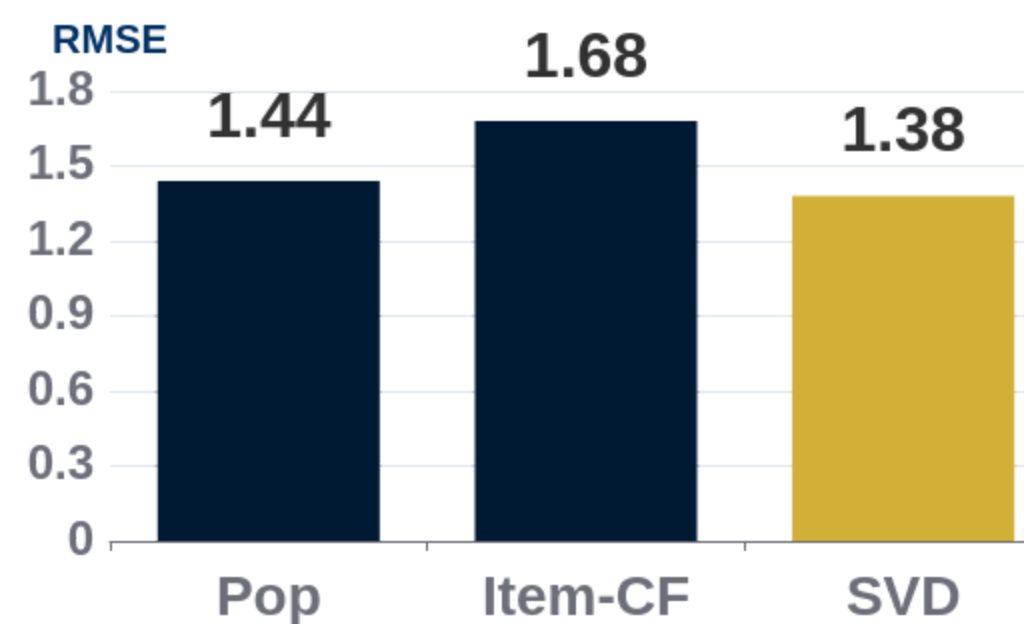
Additional Model Performance Metrics

Metric	Baseline	Item-CF	SVD
MAE	1.24	1.37	1.18★
Precision@10	0.0008	0.0006	0.0010★
Recall@10	0.0052	0.0040	0.0070★

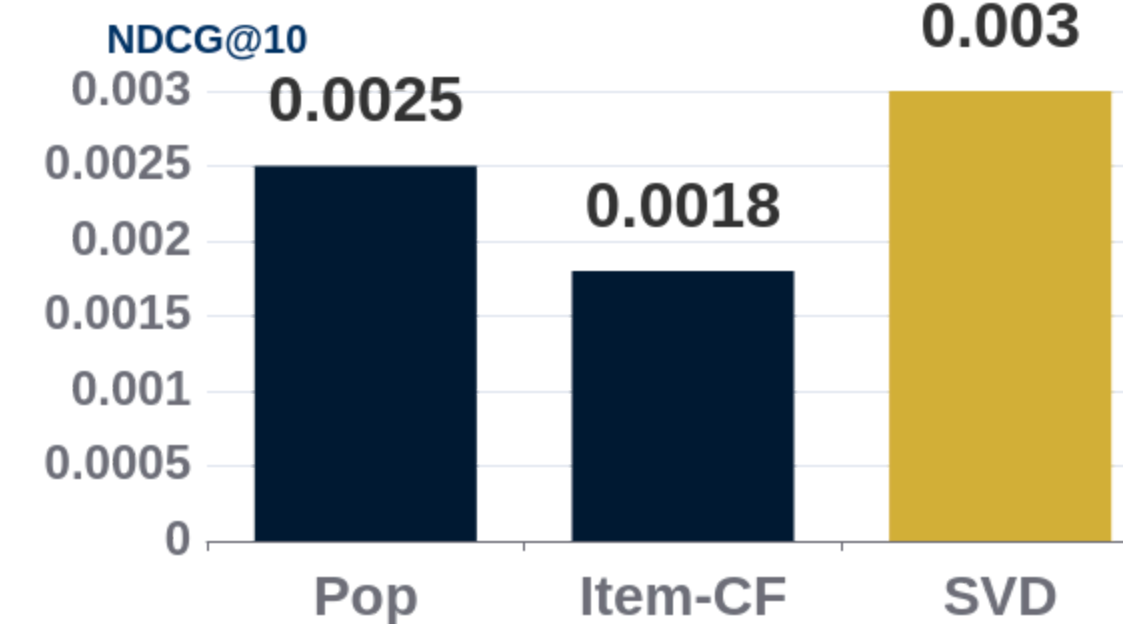
Interpretation of Results

SVD reduced prediction error compared with Item-CF while achieving the strongest ranking quality. SVD outperformed competing models in Precision@10, Recall@10, and NDCG@10.

RMSE Error



NDCG@10 Quality



6. Key Findings

1 SVD: Lowest RMSE (1.38)

2 SVD: Highest NDCG (0.0030)

3 Item-CF: Max Coverage (99.85%)

4 Pop: Best for Cold-Start

5 Sparsity (99.50%): Primary Challenge

7. Business Impact

🔍 Improved Product Discovery

👤 Enhanced Personalization

⚡ Relevance & Engagement

📦 Scalable Framework

👥 Customer Retention

8. Limitations & Future Research

Limitations

- High matrix sparsity (99.50%)
- Limited interaction overlap
- Cold-start challenges

Future Research

- Hybrid recommender systems
- Content-based filtering
- Deep-learning recommenders

RECOMMENDATIONS

Primary: SVD Matrix Factorization

Best balance of accuracy, ranking quality, and personalization.

Support: Item-Based CF

Best coverage for broader product discovery.

Fallback: Popularity Baseline

Useful for cold-start scenarios.

"SVD IS THE RECOMMENDED ENGINE FOR AMAZON BEAUTY"